

ANMOL PATALAY

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Education

Maharashtra Institute of Technology, WPU - CGPA: 9.26/10

Jul 2021 – Jul 2025

Bachelor of Technology in Electronics and Communication Engineering (AI & ML Specialization)

Pune, Maharashtra

Professional Summary

Backend Software Engineer with production experience building Python REST APIs using FastAPI, authentication/authorization systems (JWT, OAuth2), and database-backed services with MySQL, SQLAlchemy, and Alembic migrations. Comfortable working independently through unfamiliar codebases – investigating root causes, integrating third-party services, and shipping fixes to production. Background in AI/ML from applied academic and industry projects.

Experience

Visteon Corporation

August 2025 – Present

Software Engineer

Pune, Maharashtra

- Built a FastAPI backend with Python listeners to stream live execution data in real time, exposing endpoints for downstream system consumption.
- Designed modular Python libraries wrapping lower-level automation and hardware-interface functions, giving full ownership of the codebase end-to-end and enabling reuse across multiple validation features.
- Converted 652 Excel-defined manual test cases into 257 auto-generated Robot Framework test files using Python file-handling and pandas-driven parsing, eliminating manual keyword authoring at scale.
- Built a computer vision and OCR pipeline processing a 40 FPS camera feed to validate on-screen instrument cluster content in real time, replacing manual visual inspection.
- Automated CAN bus signal transmission, DBC-based signal parsing, and NVM configuration changes post security-unlock, entirely in Python; debugged low-level driver issues (Vector XL CAN interface) blocking test execution.
- Cut manual testing time by 70% by replacing repetitive validation workflows with reusable automated components across the suite.

Projects

Job Application Tracker GitHub

Jul 2026 – Aug 2026

- Built a RESTful backend with 20+ endpoints using FastAPI, SQLAlchemy ORM, MySQL, and Pydantic, organized into modular routers by resource for maintainability.
- Managed schema evolution with Alembic migrations, keeping model changes version-controlled and reversible.
- Implemented OAuth2 Password Flow (JWT) authentication with bcrypt hashing, dependency-injected authorization guards, and user-scoped access control.
- Added rate limiting on authentication endpoints to mitigate brute-force/abuse patterns, plus structured error handling and a dedicated automated test suite.

The Couples App GitHub

2026

- Built a FastAPI + MongoDB backend integrating Google Identity Services (OpenID Connect) for third-party sign-in, verifying Google ID tokens server-side alongside a native email/password flow.
- Implemented short-lived access JWTs with a separate, hashed, rotatable refresh token stored in Mongo – supporting revocation without invalidating other sessions.
- Designed a multi-stage authorization boundary via chained route guard dependencies, scoping shared resources to exactly one linked account pair.

Technical Skills

Backend/APIs: FastAPI, Flask, REST APIs, SQLAlchemy, Alembic (migrations), JWT/OAuth2, OpenID Connect, Third-Party API Integration, Rate Limiting, Redis (basics)

Languages: Python, C++, SQL

SQL/Databases: MySQL, MongoDB, Complex Joins, CTEs, Window Functions, Views, Normalization, Query Optimization, Indexing, Transactions (ACID), ORM & migration tooling

Security: Password Hashing (bcrypt), Token-Based Auth, Authorization Boundaries & Access Control, Secure Credential Handling

DevOps/Tools: Docker (multi-stage builds), Git, GitHub Actions (CI/CD), Linux CLI

AI/ML: scikit-learn, TensorFlow, PyTorch, LangChain, HuggingFace Transformers, OpenAI API